

EVIDENCE FOR THE BIG BANG

The strongest evidence for the Big Bang is the radiation it left, called the cosmic microwave background radiation (CMBR). George Gamow (see panel, opposite) predicted the radiation's existence in 1948. Its detection in the 1960s was confirmation, for most cosmologists, of the Big Bang theory. Other observations help support the theory.

BACKGROUND RADIATION The spectrum of the CMBR, discovered by Arno Penzias and Robert Wilson (below), indicates a uniformly hot early universe.



EXPANSION If the universe is expanding and cooling, it must once have been much smaller and hotter.

BALANCE OF ELEMENTS Big Bang theory exactly predicts the proportion of light elements (hydrogen, helium, and lithium) seen in the universe today.

GENERAL RELATIVITY Einstein's theory predicts that the universe must either be expanding or contracting—it cannot stay the same size.



DARK NIGHT SKY If the universe were both infinitely large and old, Earth would receive light

from every part of the night sky and it would look bright—much brighter even than the densest star field (above). The fact that it is not is called Olbers' paradox. The Big Bang resolves the paradox by proposing that the universe has not always existed.

10⁸K (180 million °F/100 million °C)

PHOTON EPOCH

During this relatively lengthy epoch, the ocean of matter particles (comprising mainly electrons, protons, and helium nuclei) were in a continual state of interaction with photons (radiant energy), making the universe "foggy."

1,000 seconds

10,000 light-years

100 million light-years

4,000K (6,740°F/3,727°C)

RECOMBINATION EPOCH

During this epoch, atoms formed as atomic nuclei combined with electrons. About nine hydrogen atoms were made for each helium atom. A few atoms of lithium and deuterium (heavy hydrogen) also formed. Photons were now free to travel through the universe.

380,000 years

